# 图片镜像旋转

图像的旋转是围绕一个特定点进行的，而图像的镜像旋转则是围绕坐标轴进行的。图像的镜像旋转分为水平翻转、垂直翻转、水平垂直翻转三种。

水平翻转就是将图片的像素点沿y轴翻转，具体到像素点来说就是令其坐标从（x,y）翻转为（-x,y）。

垂直翻转就是将图片的像素点沿x轴翻转，具体到像素点来说就是其坐标从（x,y）翻转为（x,-y）

水平垂直翻转就是水平翻转和垂直翻转的结合，具体到像素点来说就是其坐标从（x,y）翻转为（-x,-y）。用公式表示为：

$$dst_{i,j} = \begin{cases} src_{src.height-i-1,j} & flipCode = 0 \\ src_{i,src.width-j-1} & flipCode > 0 \\ src_{src.height-i-1,src.width-j-1} & flipCode < 0 \end{cases}$$

由图像的旋转我们知道，图像在旋转的时候需要有旋转中心，而图像的镜像旋转虽然都是围绕x轴和y轴进行旋转，但是我们也需要确定x轴和y轴的坐标。在OpenCV中，图片的镜像旋转是以图像的中心为原点进行镜像翻转的。也就是说，水平翻转时，图像的左侧和右侧会关于中心点进行交换，垂直翻转时，图像的上侧和下侧会关于中心点进行交换。

在本实验中，图片镜像旋转组件里的参数flipcode有三个选择：

- 0：垂直翻转
- 大于0：水平翻转
- 小于0：水平垂直翻转

## 1. 编程实例与步骤

需要通过元宇宙平台进入“人工智能虚拟仿真实验”界面和打开“华清远见人工智能虚拟仿真本地服务管理平台”软件。

### 1.1 （可选）使用“华清远见人工智能虚拟仿真本地服务管理平台”启动服务

服务启动一次之后就不需要再次启动了。

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## 1.2 单击图片镜像旋转实验进入

如果之前打开了进入了某个实验，可以点击“返回”回到主页即可。

## 1.3 拖出组件并连线

根据实验原理，将我们的逻辑抽象成组件之后，连线如下图所示。

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## 1.4 验证

点击验证，如显示校验成功，即代表逻辑无误，验证按钮与校验成功界面如下图所示。

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## 1.5 运行

点击运行，可以实现功能运行，并显示结果输出，如下图所示。

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## 1.6 代码生成

点击界面左下角的生成代码按钮，可生成本实验的Python代码，生成代码按钮如下图所示。

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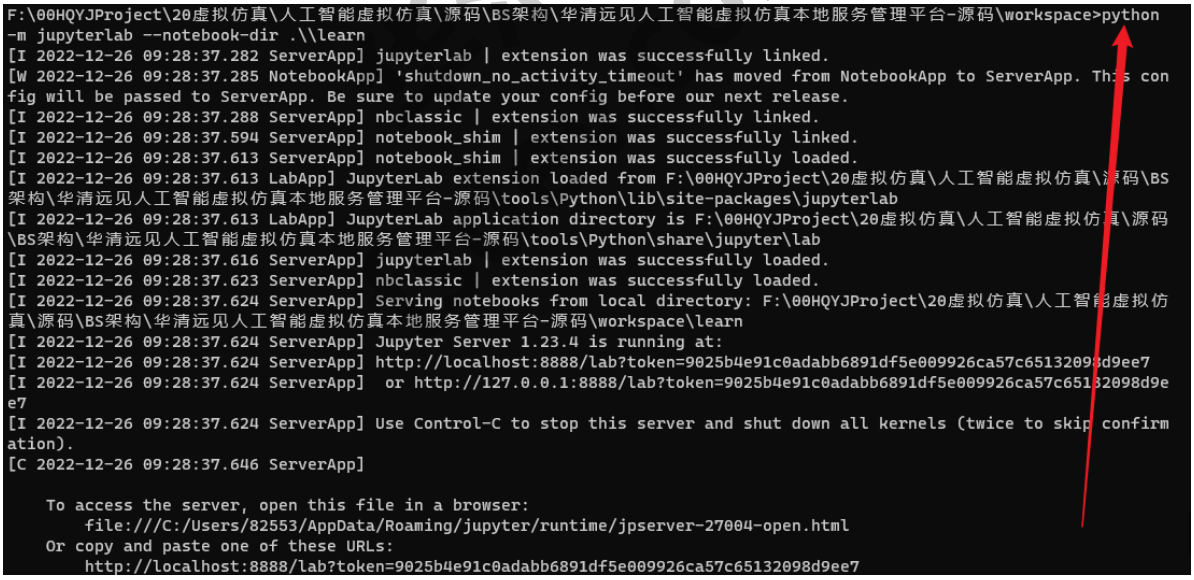
点击后可以查看该实验代码，代码会根据界面所选参数进行适配，点击复制可以将其复制，并粘贴在其它地方。

这里我们可以粘贴到软件的Jupyter中运行，“华清远见人工智能虚拟仿真本地服务管理平台”打开Jupyter，第一次打开Jupyter需要点击“打开CMD命令”，打开Jupyter后无需再次打开CMD，如下图所示。

在弹出的CMD命令行中，输入python -m jupyterlab --notebook-dir .\\learn，点击回车，浏览器会自动跳转打开Jupyter，如下图所示。



```
F:\00HQYJProject\20虚拟仿真\人工智能虚拟仿真\源码\BS架构\华清远见人工智能虚拟仿真本地服务管理平台-源码\workspace>python
-m jupyterlab --notebook-dir .\\learn
[I 2022-12-26 09:28:37.282 ServerApp] jupyterlab | extension was successfully linked.
[W 2022-12-26 09:28:37.285 NotebookApp] 'shutdown_no_activity_timeout' has moved from NotebookApp to ServerApp. This con
fig will be passed to ServerApp. Be sure to update your config before our next release.
[I 2022-12-26 09:28:37.288 ServerApp] nbclassic | extension was successfully linked.
[I 2022-12-26 09:28:37.594 ServerApp] notebook_shim | extension was successfully linked.
[I 2022-12-26 09:28:37.613 ServerApp] notebook_shim | extension was successfully loaded.
[I 2022-12-26 09:28:37.613 LabApp] JupyterLab extension loaded from F:\00HQYJProject\20虚拟仿真\人工智能虚拟仿真\源码\BS
架构\华清远见人工智能虚拟仿真本地服务管理平台-源码\tools\Python\lib\site-packages\jupyterlab
[I 2022-12-26 09:28:37.613 LabApp] JupyterLab application directory is F:\00HQYJProject\20虚拟仿真\人工智能虚拟仿真\源码
\BS架构\华清远见人工智能虚拟仿真本地服务管理平台-源码\tools\Python\share\jupyter\lab
[I 2022-12-26 09:28:37.616 ServerApp] jupyterlab | extension was successfully loaded.
[I 2022-12-26 09:28:37.623 ServerApp] nbclassic | extension was successfully loaded.
[I 2022-12-26 09:28:37.624 ServerApp] Serving notebooks from local directory: F:\00HQYJProject\20虚拟仿真\人工智能虚拟仿
真\源码\BS架构\华清远见人工智能虚拟仿真本地服务管理平台-源码\workspace\learn
[I 2022-12-26 09:28:37.624 ServerApp] Jupyter Server 1.23.4 is running at:
[I 2022-12-26 09:28:37.624 ServerApp] http://localhost:8888/lab?token=9025b4e91c0adabb6891df5e009926ca57c65132098d9ee7
[I 2022-12-26 09:28:37.624 ServerApp]  or http://127.0.0.1:8888/lab?token=9025b4e91c0adabb6891df5e009926ca57c65132098d9e
e7
[I 2022-12-26 09:28:37.624 ServerApp] Use Control-C to stop this server and shut down all kernels (twice to skip confirm
ation).
[C 2022-12-26 09:28:37.646 ServerApp]

    To access the server, open this file in a browser:
        file:///C:/Users/82553/AppData/Roaming/jupyter/runtime/jpserver-27004-open.html
    Or copy and paste one of these URLs:
        http://localhost:8888/lab?token=9025b4e91c0adabb6891df5e009926ca57c65132098d9ee7
```

如果没有自动跳转和打开浏览器，请输入下图中圈出的内容，每个人的该内容不一致，请复制自己的内容。

```
F:\00HQYJProject\20虚拟仿真\人工智能虚拟仿真\源码\BS架构\华清远见人工智能虚拟仿真本地服务管理平台-源码\workspace>python
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架构\华清远见人工智能虚拟仿真本地服务管理平台-源码\tools\Python\lib\site-packages\jupyterlab
[I 2022-12-26 09:28:37.613 LabApp] JupyterLab application directory is F:\00HQYJProject\20虚拟仿真\人工智能虚拟仿真\源码
\BS架构\华清远见人工智能虚拟仿真本地服务管理平台-源码\tools\Python\share\jupyter\lab
[I 2022-12-26 09:28:37.616 ServerApp] jupyterlab | extension was successfully loaded.
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真\源码\BS架构\华清远见人工智能虚拟仿真本地服务管理平台-源码\workspace\learn
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    Or copy and paste one of these URLs:
        http://localhost:8888/lab?token=9025b4e91c0adabb6891df5e009926ca57c65132098d9ee7
```

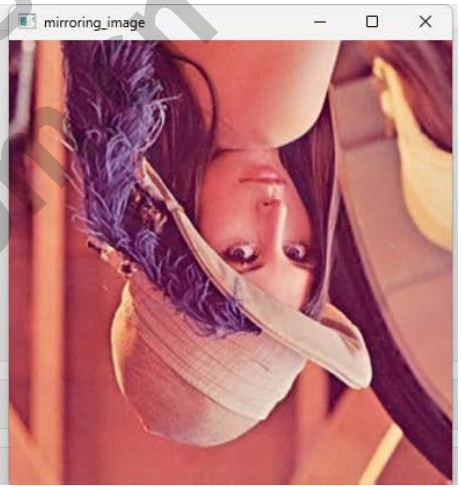
将代码输入到Jupyter中，注意这里需要往Jupyter导入一张demo.png，在大多数情况下，这张照片最好不要是其它后缀修改成png后缀而来，而是直接后缀就是png。点击运行可以查看结果。

## 2. 实验现象

可以看到，图片镜像旋转组件中不同的参数对于图片镜像旋转后的结果不同，实验结果为：

```
import cv2

if __name__ == "__main__":
    path = "./images/lena.png"
    image_np = cv2.imread(path)
    # 对图片进行镜像操作
    mirroring_image = cv2.flip(image_np, 0)
    cv2.imshow("mirroring_image", mirroring_image)
    cv2.waitKey(0)
```

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